

Speech Processing Mini Report

This report describes a basic speech processing pipeline including waveform analysis, time–frequency representations, Mel features, MFCCs, and a keyword spotting model.

Speech as a Signal

Speech is a non-stationary time-domain signal. Voiced sounds show periodicity, while unvoiced sounds resemble noise. Spectral features capture formant structure.

Feature Extraction

We use STFT-based spectrograms, Mel spectrograms aligned with human perception, and MFCCs obtained via DCT of log-Mel energies.

Keyword Spotting Model

A CNN classifier is trained on Mel spectrogram inputs to distinguish between keywords such as 'yes' and 'no'. This demonstrates a neural approach compared to classical GMM-HMM pipelines.

Conclusion

This mini-project integrates signal processing intuition with deep learning, providing a foundation for modern speech recognition systems.